EXPERIMENT : 2

Analyze the data and perform data preprocessing and visualization.

AIM:

Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

DESCRIPTION:

Use sample data set sales-data.csv.

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the data
file_path = '/content/drive/MyDrive/Colab Notebooks/SALES DATA.csv'
df = pd.read_csv(file_path)

# Display first few rows
print("First few rows:")
print(df.head())

# Check missing values
print("\nMissing values:")
print(df.isnull().sum())
```

```
# Fill missing Sales values with mean
df['Sales'] = df['Sales'].fillna(df['Sales'].mean())

# Remove rows with missing values
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)

# Summary statistics
print("\nSummary Statistics:")
print(df.describe())

# Group by Product
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()

print("\nProduct Summary:")
print(product_summary)

# Bar plot
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.xticks(rotation=45)
plt.show()

df['Date'] = pd.to_datetime(df['Date'], errors='coerce')
```

```
# Remove invalid dates
df.dropna(subset=['Date'], inplace=True)

# Group sales by date
sales_over_time = df.groupby('Date').agg({
    'Sales': 'sum'
}).reset_index()

# Line plot
plt.figure(figsize=(10, 6))
plt.plot(
    sales_over_time['Date'],
    sales_over_time['Sales']
)

plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.xticks(rotation=45)
plt.show()

pivot_table = df.pivot_table(
    values='Sales',
    index='Region',
    columns='Product',
    aggfunc='sum',
    fill_value=0
)

print("\nPivot Table:")
print(pivot_table)
```

```
numeric_df = df.select_dtypes(include=np.number)

correlation_matrix = numeric_df.corr()

print("\nCorrelation Matrix:")
print(correlation_matrix)



plt.figure(figsize=(8, 6))

sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap='coolwarm'
)

plt.title('Correlation Matrix')
plt.show()
```

OUTPUT:

First few rows:

	Date	Product	Sales	Quantity	Region
0	01-01-2023	Product A	200	4	North
1	02-01-2023	Product B	150	3	South
2	03-01-2023	Product A	220	5	North
3	04-01-2023	Product C	300	6	East
4	05-01-2023	Product B	180	4	West

Missing values:

Date	0
Product	0
Sales	0
Quantity	0

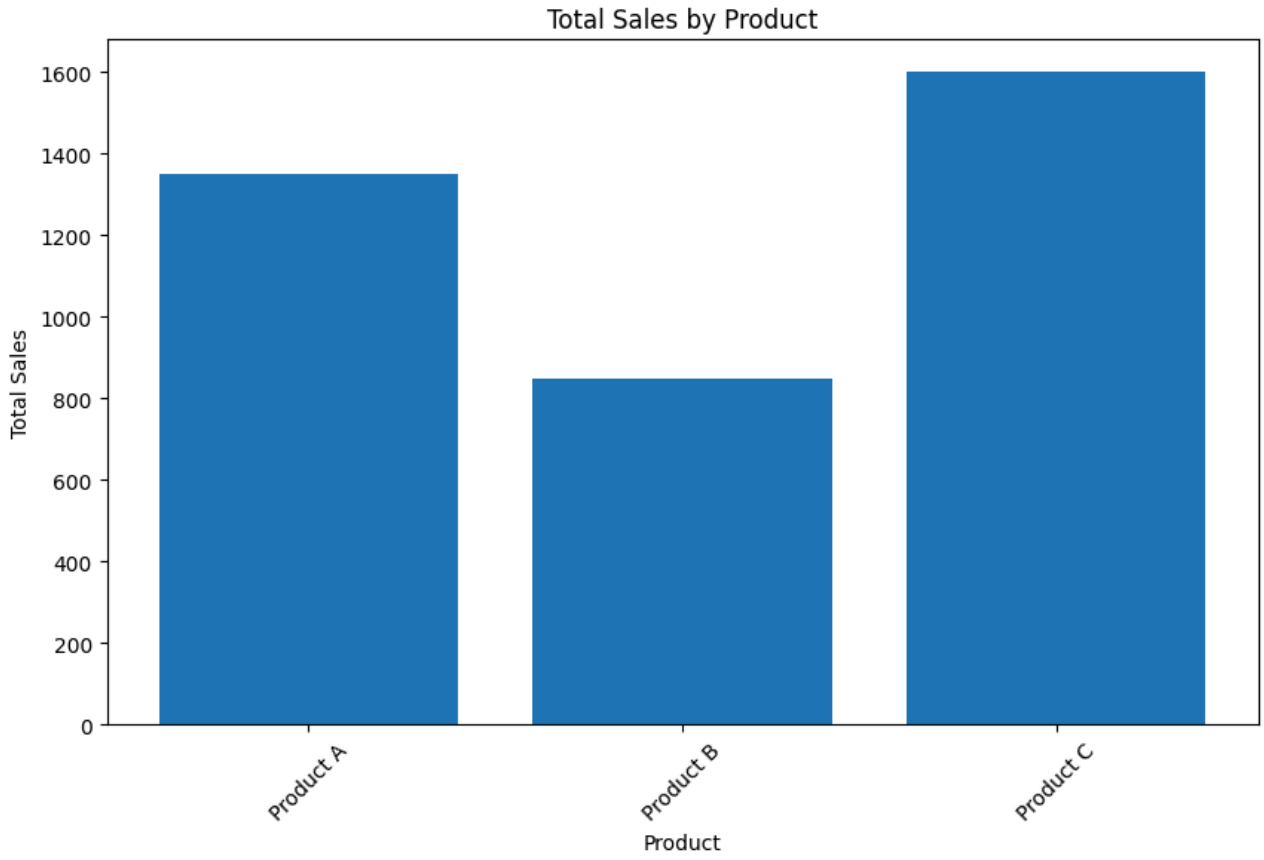
Region 0
dtype: int64

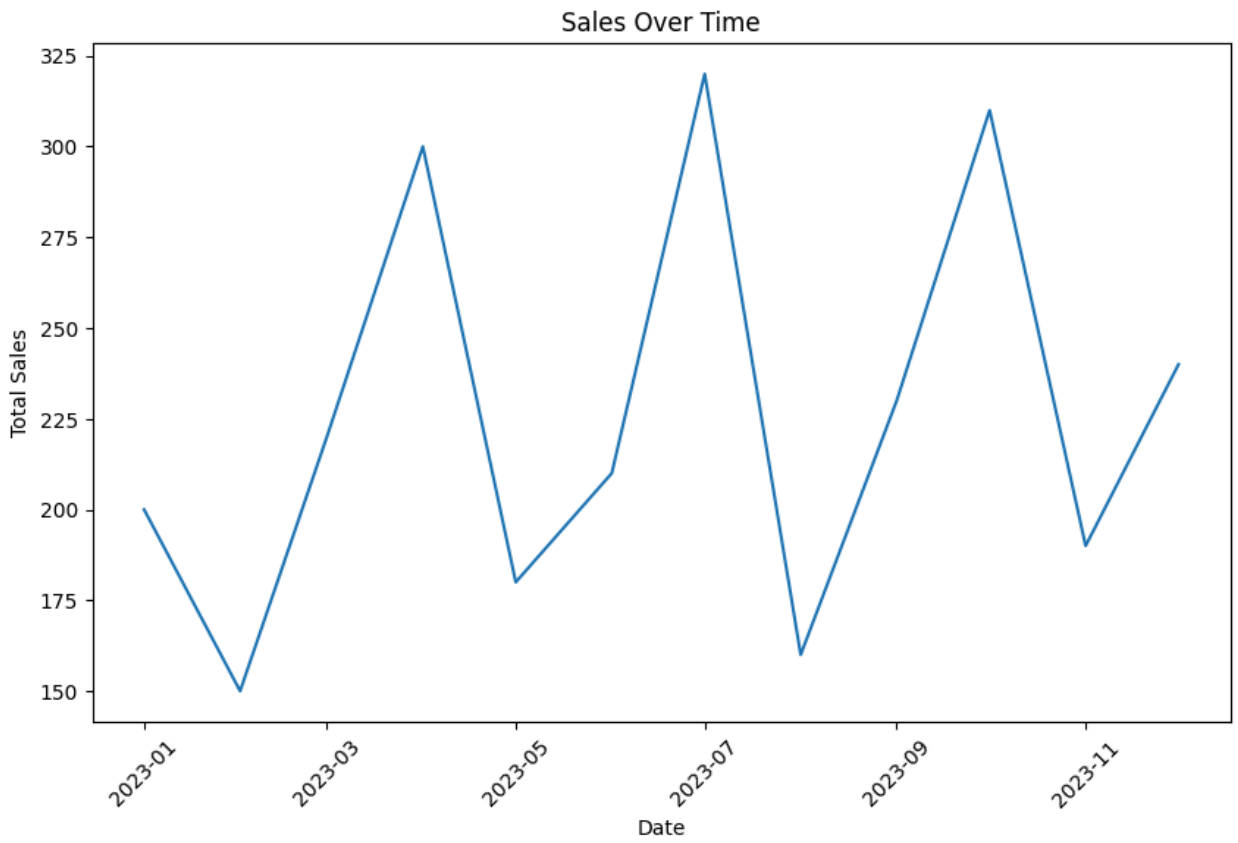
Summary Statistics:

	Sales	Quantity
count	16.000000	16.000000
mean	237.500000	5.375000
std	64.031242	1.746425
min	150.000000	3.000000
25%	187.500000	4.000000
50%	225.000000	5.500000
75%	302.500000	7.000000
max	340.000000	8.000000

Product Summary:

	Product	Sales	Quantity
0	Product A	1350	33
1	Product B	850	17
2	Product C	1600	36



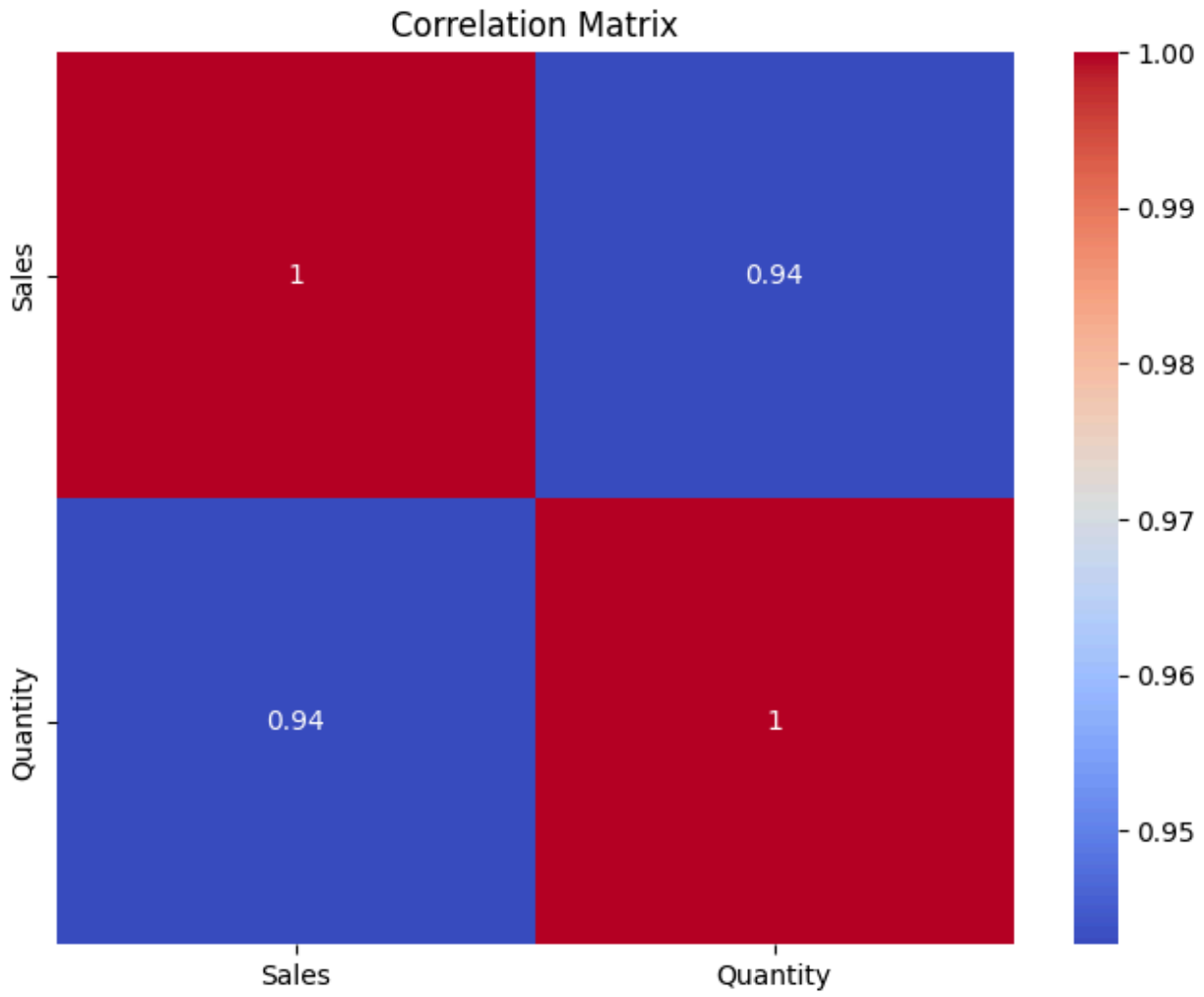


Pivot Table:

Product	Product A	Product B	Product C
Region			
East	0	0	930
North	1100	0	0
South	0	310	0
West	0	370	0

Correlation Matrix:

	Sales	Quantity
Sales	1.00000	0.94268
Quantity	0.94268	1.00000



RESULT:

Thus the given csv dataset was successfully uploaded,analyzed and preprocessed. The data was visualized using different plots and the required information was acquired